

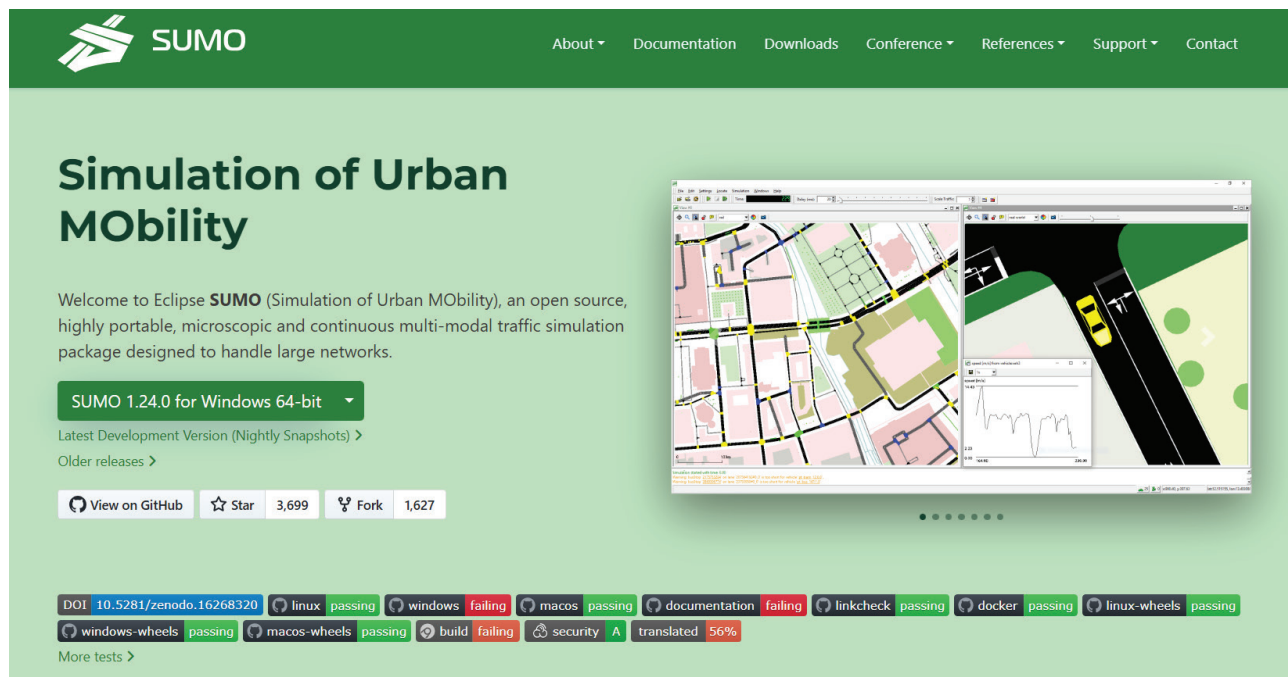


Figure 3: SUMO simulator for urban mobility and vehicular networks

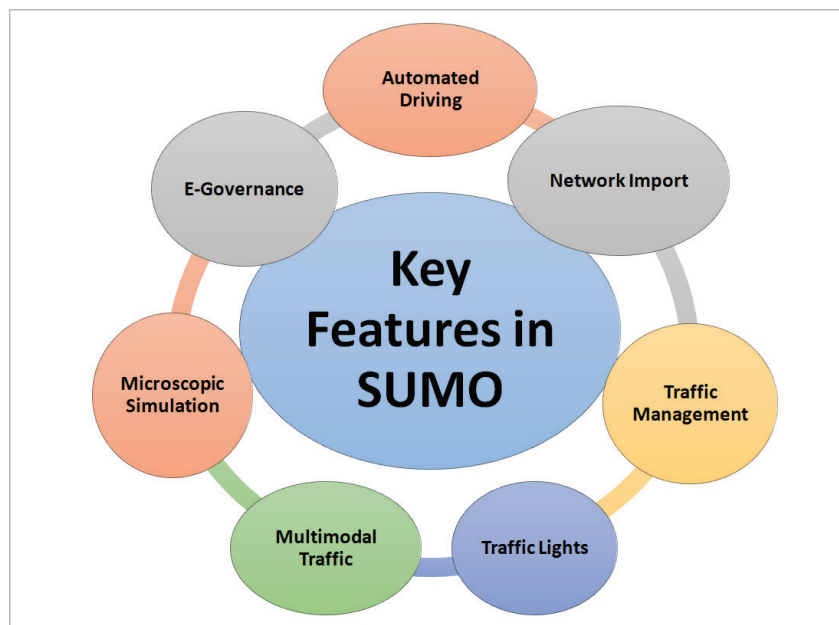


Figure 4: Key features of SUMO

vehicular communications. Multiple vehicles of different types can be created.

sumogui: This is a GUI-based working panel for running and visualising the simulations associated with urban mobility and vehicular communications.

netconvert: netconvert is used to import the road networks from various sources for a real-time look at the implementation scenarios.

od2trips: od2trips is used to import the origin-destination traffic along with zone information. It helps analyse the

simulation from multiple perspectives.

Here's how you can create a simulation scenario using SUMO.

- Open *netedit* to design a road network with vehicular information.
- The network scenario can be created from the interface available at osmWebWizard.
- Traffic demand is implemented with a route *xml* file that specifies the types of vehicles, multiple routes, departure time and related information.
- The GUI of Sumo is launched by loading the *sumocfg* file to visualise the simulation.

The Open Street Map is used with osmWebWizard so that a real-time map of the city or specific location can be accessed. SUMO modules integrate random traffic with real-time roads so that simulations using various algorithms can be done effectively.

Once the scenario is created on SUMO, there are multiple options to simulate the designed network on different types of interfaces.

Researchers continue to experiment with the simulation of smart cities and smart vehicular networks on a wide